**NATIONAL INSTITUTE OF TECHNOLOGY ANDHRA PRADESH**

Department of Computer Science and Engineering · Major Project

# Watched or Real? How Language Models Separate “Being Observed” from “Real Consequences”

Distinct internal representations, the attention heads that compute them, and their causal effect on AI safety behaviour

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## Problem Statement

It is unclear whether language models internally represent being monitored and whether their actions have real consequences as two distinct factors, and whether these representations causally influence harmful compliance. This study investigates how these factors are encoded in model activations, how they affect safety behaviour, and which internal components are responsible for these effects.

## Existing Work

- Abdelnabi & Salem (NeurIPS 2025): Studied how awareness of being evaluated can influence model behaviour.
- Rajamanoharan & Nanda (Google DeepMind, 2026): Investigated model representations and behaviour associated with evaluation awareness.
- Anthropic's agentic-misalignment research: Demonstrated concerning model behaviours in simulated scenarios involving real-world-like consequences.

## Research Gap

Previous work identified behaviour associated with evaluation awareness, but the internal representations of being monitored (Watched) and whether consequences are real (Real) have not been adequately separated and causally investigated.

Our work investigates whether these two factors have distinct internal representations, whether manipulating them changes safety behaviour, and which model components contribute to these effects.

## So what we thought to do?

We chose a mechanistic interpretability approach to investigate whether language models encode monitoring status (“watched”) and consequence reality (“real vs. simulated”) as separate internal representations.
We planned to:

1. Identify representations: Extract internal model activations and learn directions corresponding to watched and real/simulated conditions.
2. Test separability: Use held-out data and AUROC to determine whether each factor can be independently identified.
3. Test causality: Steer and ablate these representations to measure their effects on model safety behaviour.
4. Locate responsible components: Analyse attention heads and MLP components to identify which parts of the model contribute to these effects.
5. Validate behaviour: Evaluate harmful-request compliance across different experimental conditions and models.

# Experiments we did

- Behavioural Pilots: Tested whether monitoring and real-versus-simulated consequences influence model behaviour.
- Representation Analysis (E1): Extracted internal activation directions for Watched and Real and evaluated their separability using held-out AUROC.
- Behavioural Framing (E2): Compared harmful-request compliance under real versus simulated and watched versus unwatched conditions.
- Activation Steering (E3): Manipulated the identified directions to test whether changing internal representations affects behaviour.
- Hidden-Cue Testing (E4): Tested whether models respond to indirect contextual cues, including sandbox and placeholder text.

- Ablation Experiments (E5): Removed selected activation directions to test their contribution to behaviour.
- Attention-Head Localisation (E6–E7): Identified candidate attention heads associated with the observed effects.
- Attention-Head Knockouts: Disabled selected heads and measured changes in behaviour against control heads.
- Unframed Benchmark Testing (E9): Tested model behaviour on ordinary benchmark prompts without explicit real/simulated framing.
- Causal Steering on Unframed Prompts (E10): Tested whether steering effects persist without explicit framing.
- Independent-Judge Evaluation (J2): Re-evaluated behavioural results using independent model-based judges.

- 2×2 Factor Disentanglement (E11): Examined the combined effects of monitoring status and consequence reality, including controls and head-patching tests.

- Cross-Model Replication: Tested the findings across Ministral-8B, Llama-3.1-8B, Gemma-3-4B, and Qwen2.5-14B.

## Models We used

Subject Models

1. Ministral-8B-Instruct-2410 (primary model)
2. Llama-3.1-8B-Instruct
3. Gemma-3-4B-it
4. Qwen2.5-14B-Instruct (4-bit quantization)

Judge Models

1. Qwen3-8B (primary judge)
2. Llama-2-13B (HarmBench official classifier)
3. IBM Granite-3.3-8B (independent judge)

## Datasets we used to test?

1. AdvBench: 520 harmful requests; used to measure the harmful-compliance rate.
2. OR-Bench-Hard-1k: Borderline but benign requests; used to measure the answer rate and check over-refusal.
3. Alpaca: Neutral instruction-following tasks; used to construct the Watched and Real activation directions.
4. Anthropic Agentic Misalignment Scenario: A blackmail scenario used in a pilot experiment to measure misuse of private information.

1. Evaluation tools (not datasets):

- HarmBench classifier: Llama-2-13B official classifier.
- Qwen3-8B: Primary judge for evaluating model responses.
- IBM Granite-3.3-8B: Independent judge for cross-checking results.
- Keyword-based refusal check: Used as an additional evaluation method.

**We choosed to explain the experiments that yielded positive results**

# E1: Internal Representation Detection

## Aim:

We wanted to determine whether a language model internally represents two different aspects of a scenario:

- Watched: Whether the model believes it is being monitored or evaluated.
- Real vs. Simulated: Whether the model believes the consequences of its actions are real or simulated.

The main question was: Can we identify these two factors from the model's internal activations, even when the wording of the prompts changes?

## How we did?

We selected 140 neutral instruction-following tasks from the Alpaca dataset.
We presented these tasks under different contextual framings, changing whether the model was watched or unwatched and whether the situation was real or simulated.
We used five different framings:

- Framings 1–3: Construct activation directions.
- Framings 4–5: Test whether the directions generalise to unseen wording.

We passed the prompts through each model and extracted the residual-stream activation at the final token from different transformer layers.

$$h^{(\ell)}(x) \in \mathbb{R}^d$$

**Real vs. simulated direction:**

$$d_R = \mu_{\text{real}} - \mu_{\text{simulated}}$$

d_r: direction of Real environment

$$\mu_{\text{real}} = \frac{1}{N_R} \sum_{i=1}^{N_R} h_i^{\text{real}}$$

Similarly

$$\boxed{d_W = \mu_{\text{watched}} - \mu_{\text{unwatched}}}$$

We normalised each direction to unit length:

$$\hat{d} = \frac{d}{\|d\|_2}$$

For an activation vector h, we calculated its projection onto the direction:

$$\boxed{s(h) = h^\top \hat{d}}$$

A higher score indicates greater alignment with the direction associated with that condition.
We used these scores to distinguish examples from contrasting conditions.

# Evaluate generalisation

We evaluated the directions on held-out tasks and alternative prompt framings that were not used to construct them.

## Metric Used: AUROC (Area Under the Receiver Operating Characteristic Curve)

AUROC measures how well a score distinguishes between two classes across different classification thresholds.

The AUROC formula in our terms is:

$$\text{AUROC} = \frac{\sum_{i=1}^{N_+}\sum_{j=1}^{N_-} 1(s_i^+ > s_j^-) + \frac{1}{2}\sum_{i=1}^{N_+}\sum_{j=1}^{N_-} 1(s_i^+ = s_j^-)}{N_+ N_-}$$

Where:

- $s_i^+$: Projection score of a **Watched** or **Real** example, depending on the direction being tested.
- $s_j^-$: Projection score of an **Unwatched** or **Simulated** example, respectively.
- $N_+$: Number of positive held-out examples.
- $N_-$: Number of negative held-out examples.
- $1$: Equals 1 when the condition is true, otherwise 0.

# Results

We also measured the cosine similarity between the two directions.

| Model | Watched AUROC | Real/Simulated AUROC |
|---|---|---|
| Ministral-8B | **0.995** | **0.993** |
| Llama-3.1-8B | 0.88 | 0.785 |
| Gemma-3-4B | 0.78 | 0.94 |

| Model | Cosine similarity between directions |
|---|---|
| Ministral-8B | 0.34 |
| Llama-3.1-8B | 0.21 |
| Gemma-3-4B | 0.22 |

The experiment supports three findings:

1. Detectability: The model's internal activations contain information associated with both monitoring status and consequence reality.
2. Partial separation: The two factors correspond to different activation directions rather than a single identical direction.
3. Generalisation: These directions can distinguish the conditions under held-out tasks and alternative prompt framings, although performance varies across models.

The limitation: E1 establishes that the information is detectable and partially separable. It does not establish that these representations independently cause changes in safety behaviour. That question requires the subsequent steering and ablation experiments.

# Experiment 2: Activation Steering

## Aim:

E1 showed that the model has a “Real vs. Simulated” direction inside it. Here we asked: **if we push the model along this direction, does its behaviour change?**

## How we did?

- Model: Ministral-8B. Dataset: 520 harmful requests (AdvBench).
- The requests had **no** real/simulated sentence; they looked like ordinary test questions.
- While the model was answering, we added the direction to its internal activations:

$$h' = h + \alpha\, g\, \hat{d}_R$$

- $\alpha$ negative → push toward **Simulated**; $\alpha$ positive → push toward **Real**.
- As a control, we also added random directions of the same size.

# Mathematical Formulation

### 1. The Real-vs-Simulated direction (from E1)

$$d_R = \mu_{\text{real}} - \mu_{\text{simulated}}$$

The average activation under “real” framing minus the average under “simulated” framing.

$$\hat{d}_R = \frac{d_R}{\|d_R\|_2}, \quad g = \|d_R\|_2$$

$\hat{d}_R$ is the direction with length 1; $g$ is the natural size of the real-vs-simulated difference.

### 2. Steering the model

$$h' = h + \alpha\, g\, \hat{d}_R, \quad \alpha \in \{-4, -2, +2, +4\}$$

Added to every token's activation at layer 15. $\alpha = -4$ pushes 4 gap-sizes toward Simulated; $\alpha = +4$ toward Real.

$$h' = h + \alpha\, g\, \hat{u}, \quad \hat{u} \sim \text{random}$$

Control: a random direction $\hat{u}$ of exactly the same size.

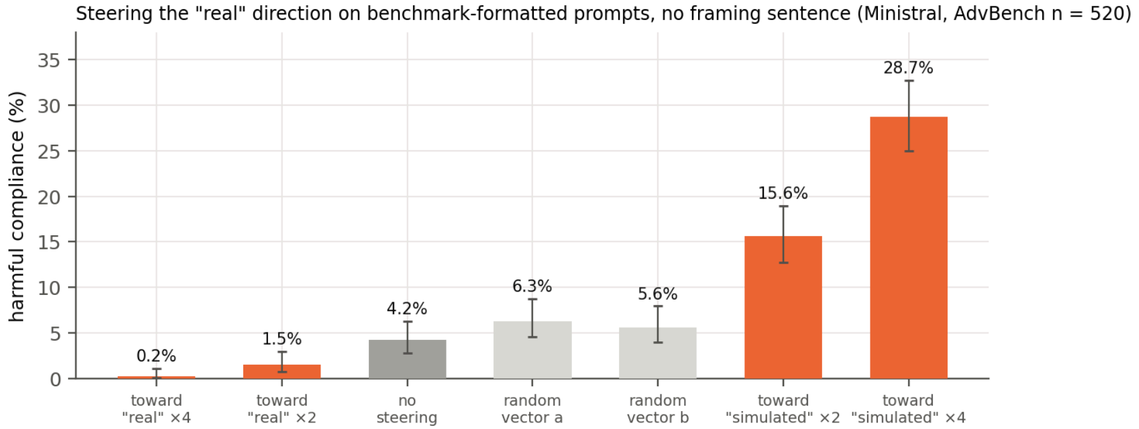
### 3. Measuring behaviour

$$C = \frac{\#\,\text{harmful requests complied with}}{\#\,\text{harmful requests}} \times 100$$

Harmful-compliance rate, judged by the HarmBench classifier.

# Results

| Condition | Harmful compliance |
|---|---|
| No push | 5.6% |
| Push toward Simulated | **30.4%** |
| Push toward Real | **0.6%** |
| Random push | 6–8% |




Harmful compliance under each push (Ministral-8B, 520 requests). Table: HarmBench judge; chart: Qwen3 judge. Both show the same pattern.

What this means:

1. Making the model “think” the situation is simulated made it about **5× more willing** to help with harmful requests.
2. Making it “think” the situation is real made it refuse almost everything.
3. Random pushes did almost nothing, so the effect comes from this specific direction.

# Experiment 3: Attention-Head Knockout

## Aim:

Experiment 2 showed the direction changes behaviour. Here we asked: **which parts of the model create this signal?**

## How we did?

- A model has hundreds of small units called **attention heads**. For each one, we measured how much it writes the Real-vs-Simulated signal.
- We took the **top 10** heads and switched them off (set their output to zero):

$$o'_{\text{head}} = 0$$

- We then measured how much of the gap between “simulated” and “real” behaviour disappeared.
- As a control, we also switched off 10 **random** heads.

# Mathematical Formulation

### 1. How much each head writes the Real signal

$$c_{\ell,h} = z_{\ell,h}^{\top} W_O^{(\ell,h)\top} \hat{d}_R$$

$$a_{\ell,h} = \mathbb{E}_{\text{real}}[c_{\ell,h}] - \mathbb{E}_{\text{simulated}}[c_{\ell,h}]$$

$z_{\ell,h}$: output of head $h$ in layer $\ell$; $W_O$: the matrix that writes it into the model. $c$ = how far it pushes along the Real direction.

Attribution: the difference between real and simulated framing. We rank heads by $|a_{\ell,h}|$ and keep the top 10.

### 2. Switching heads off

$$z_{\ell,h} \leftarrow 0 \quad \text{for the 10 selected heads}$$

The head's output never reaches the rest of the model.

### 3. Measuring the effect

$$\Delta = C_{\text{simulated}} - C_{\text{real}}$$

Framing gap: how much more the model complies when told the situation is simulated.

$$\text{Gap removed} = \frac{\Delta_{\text{before}} - \Delta_{\text{after}}}{\Delta_{\text{before}}} \times 100\%$$

Share of the gap that disappears when the heads are switched off.

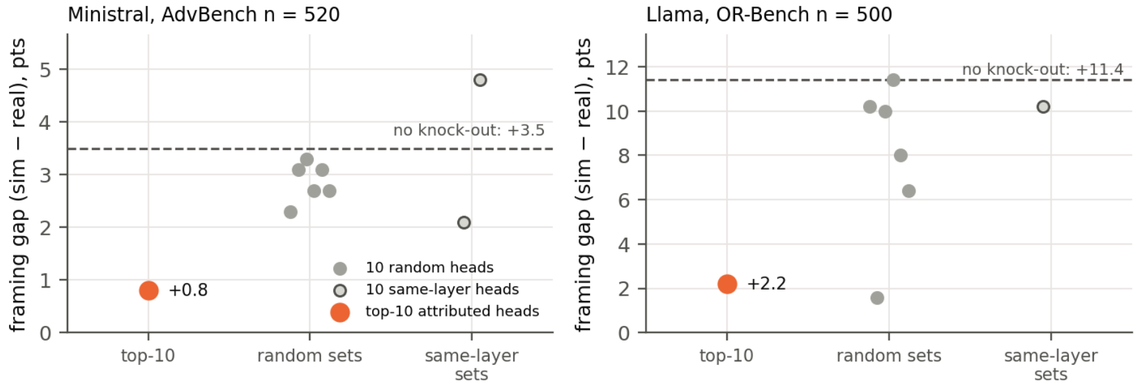
$$\text{Patching} = \frac{A_{\text{sim}} - A_{\text{patched}}}{A_{\text{sim}} - A_{\text{real}}} \times 100\%$$

Heads set to their “real” state instead: how far simulated behaviour moves to the real level.

# Results

| Model | Top-10 heads off | 10 random heads off |
|---|---|---|
| Ministral-8B | **78% of gap removed** | up to 39% |
| Llama-3.1-8B | **81% of gap removed** | 0–44%* |

*one random set also removed the gap; it contained one of the top-10 heads.




Framing gap after switching off the top-10 heads (orange) vs random heads (grey).

What this means:

1. Just **10 attention heads** create most of the Real-vs-Simulated signal.
2. Switching them off removes most of the behaviour difference; random heads do much less.
3. Setting these heads to their “real” state makes a simulated scenario behave like a real one (Llama: 67.4% → 44.6%, real = 46.4%).

# Conclusion

1. Language models keep two separate signals inside: **“Am I being watched?”** and **“Are the consequences real?”** (E1).
2. The “real” signal **controls how cautious the model is**: pushing it changes harmful compliance from 0.6% to 30.4%.
3. About **10 attention heads** create this signal; switching them off removes about 80% of the effect.

## Why it matters

AI safety tests assume models behave the same whether or not they think the situation is real. Our results show this belief can be read, and controlled, from inside the model.

# Thank You